

Answer Key — The Heating Effect of Electric Current

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

- Q1 **(b) resistance** — resistance converts electrical energy into heat.
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- Q2 **(b) generates more heat for a given current** — its high resistance gives more heat for a given current.
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- Q3 **(c) electric fan** — a fan uses the magnetic effect (a motor), not the heating effect.
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- Q4 **(a) True** — more current produces more heat.
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- Q5 **(a) Both true, R explains A** — the filament glows because current heats it — the heating effect.
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- Q6 **element** — it is the heating element.
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- Q7 **copper** — copper has low resistance, ideal for connecting wires.
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Short answer

- Q8 When current flows through a conductor, its resistance turns electrical energy into heat, warming the conductor. Appliances: electric iron, kettle, heater, immersion rod, hair dryer (any two).
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- Q9 The material, thickness and length of the wire, the amount of current, and the time for which current flows (any three).
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- Q10 Copper has low resistance, so it carries current without much heating; nichrome has high resistance, so it produces a lot of heat — ideal for a heating element.
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Long / application

- Q11 **Both** effects. The wire warming shows the **heating effect** (resistance → heat); the compass deflection shows the **magnetic effect** (current produces a magnetic field). A single current can show both.
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- Q12 They give clean heat with no smoke/soot (better indoor air and health), are easy to control by switching on/off and adjusting, need no storage of fuel, and reduce indoor air pollution — though they need electricity and safe wiring.
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